

UEHI Scoring Formula

Novelty Justification for the Urban Ecosystem Health Index

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1. Purpose of This Document

This document lays out the mathematical formula used to compute the Urban Ecosystem Health Index (UEHI) and explains what distinguishes it from a standard fixed-weight environmental index. It is written for review of the formula's novelty.

2. The Baseline Approach (Standard Practice)

Most existing environmental health indices combine multiple factors using a fixed weighted sum:

$$\text{Index} = w1.x1 + w2.x2 + \dots + wn.xn, \text{ where the sum of all } wi = 1$$

The weights (w_i) are assigned once by expert judgment and applied uniformly across every location, regardless of local conditions. This is simple and interpretable, but has three limitations this project addresses directly:

- Weights are subjective and hard to defend rigorously.
- Factors are treated as independent, so real ecological interactions between them are lost.
- The same formula is applied everywhere, even though different cities have different environmental priorities.

3. What Is Different in the UEHI Formula

3.1 Entropy-based dynamic weighting

Instead of assigning fixed weights, the UEHI computes weights from the actual variance of each parameter across the city blocks being scored. Parameters that vary more across blocks carry more information for distinguishing block health, and are weighted higher automatically.

Step 1: Normalize parameter j across all n blocks so values sum to 1:

$$p(i,j) = x(i,j) / \text{sum over } i \text{ of } x(i,j)$$

Step 2: Compute the Shannon entropy of parameter j :

$$e(j) = -k * \text{sum over } i \text{ of } [p(i,j) * \ln(p(i,j))], \text{ where } k = 1 / \ln(n)$$

Step 3: Convert entropy to a diversity score:

$$d(j) = 1 - e(j)$$

Step 4: Normalize into a final weight:

$$w(j) = d(j) / \text{sum over all parameters of } d(j)$$

This makes the weighting scheme data-driven and location-adaptive: the same 8 parameters produce a different weight distribution in a dense urban ward than in a green suburban ward, because the underlying variance differs.

3.2 Canopy-temperature synergy term

A plain weighted sum assumes every factor contributes independently. In reality, canopy cover and local temperature are causally linked: denser tree cover directly lowers surface temperature through shading and

evapotranspiration. The UEHI adds a synergy term that rewards blocks where this cooling relationship is actually observed in the data:

$$s(i) = \alpha * N(NDVI_i) * N(Temp_i), \quad \alpha = 0.1$$

Because temperature is a negative indicator, $N(Temp_i)$ is already inverted, so this term is highest when canopy is high and heat is low, meaning cooling is measurably working in that block.

3.3 Population impact multiplier

Standard indices treat population exposure as just another additive term with a fixed weight. The UEHI instead uses population as a multiplier on the environmental score, since a block's condition matters more to the NGO and to donors when more people are affected by it:

$$m(i) = 1 + \beta * N(Pop_i), \quad \beta = 0.3$$

A block with negligible population gets a multiplier close to 1 (no change). A block with maximum population in the dataset gets its score scaled up by up to 30 percent, reflecting real human stakes rather than treating an empty block and a crowded block as equally important.

4. Full Formula

With Population Exposure removed from the additive sum and used as a multiplier instead, 7 parameters remain in the entropy-weighted base score.

Normalization (applied to each of the 7 base parameters):

$$\text{Positive indicators: } N(x) = (x - x_{\min}) / (x_{\max} - x_{\min})$$

$$\text{Negative indicators: } N(x) = (x_{\max} - x) / (x_{\max} - x_{\min})$$

Base score for block i:

$$B(i) = \text{sum over } j = 1 \text{ to } 7 \text{ of } [w(j) * N(x(i,j))]]$$

Final UEHI score for block i:

$$UEHI(i) = 100 * [B(i) + s(i)] * m(i)$$

The result is capped at 100 in implementation, since the synergy term and multiplier can push it slightly above that bound.

5. Comparison Summary

Aspect	Standard fixed-weight index	UEHI (this project)
Weight assignment	Fixed by expert judgment, same for every location	Computed from Shannon entropy of the actual block data, recalculated per city
Factor interaction	Factors treated as fully independent, summed linearly	Canopy-temperature synergy term added to capture cooling interaction
Population role	Population is one more additive factor among equals	Population acts as a multiplier, scaling the score by human impact
Adaptability	Same formula applied everywhere regardless of local conditions	Weights shift automatically based on which factors actually vary in that city's data

6. Data Requirements per Parameter

For reference, the table below lists the exact data source behind each parameter in the formula, so it can be cross-checked against existing records or monitoring systems.

Parameter	Type	Source data	Role in formula
NDVI (Canopy health)	Positive	Sentinel-2 bands B8, B4 (Earth Engine)	Entropy-weighted base score
Tree Density	Positive	Segmentation output, canopy pixels per block area	Entropy-weighted base score
Biodiversity	Positive	iNaturalist / GBIF species occurrence	Entropy-weighted base score
AQI	Negative	OpenWeather Air Pollution API / CPCB stations	Entropy-weighted base score
Local Temperature	Negative	Landsat 8 Band 10, Land Surface Temperature	Entropy-weighted base score + synergy term
Impervious Surfaces	Negative	Segmentation output, built-up pixel percentage	Entropy-weighted base score
Water Availability	Positive	CGWB groundwater data / OSM distance proxy	Entropy-weighted base score
Population Exposure	Positive	WorldPop / Census 2011 gridded data	Multiplier, not additive

7. Summary Statement

The UEHI is not a relabeled weighted average. Its weights are computed from real data variance using Shannon entropy rather than assumed upfront, it includes an interaction term that models a known ecological relationship between canopy cover and temperature, and it treats population exposure as a multiplier reflecting human stakes rather than as one more additive term. Together these three changes make the index adaptive to local conditions and more representative of actual ecological and social impact than a standard fixed-weight formula.